QUESTION 1 [24 marks]

The system

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} dx/dt \\ dy/dt \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

has the general solution

$$\begin{bmatrix} x \\ y \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + \beta \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} t + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} e^{-t}$$

where $\alpha, \beta \in \mathbb{R}$.

(a) [2] eigenvalues are real & repeated $\lambda_{1,2} = -1$
one linearly independent eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(b) [1] Set $y = x'$. [1M]Thus $y' = x'' = -x - 2y$. [1A]This leads to $x'' + 2x' + x = 0$.

(c)

- [1working+2 for sketch (straight line +1direction]1 straight line orbit from $\beta = 0$
$$y = -x$$

- [3sketch = 1shape max/min +1direction+1asymptotic along e-vector] at least 4 other orbits and their directions, showing the asymptotic behaviour as
[1]End phase: As t → ∞, orbits approach (0,0) asymptotic to $x' = -x$
[1]Start phase: As t → −∞, orbits are parallel to $x' = -x$

- the slopes at which the orbits cross the x– and y– axes

[1M]slope = $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-x-2y}{y}$

[1M]along x-axis: y=0. Thus $\frac{dy}{dx}$ $is\ undefined$: slope is vertical.

[1M]along y-axis: x=0. Thus slope = -2.

- the loci of all minima and maxima in x, and also in y, if any of these exist;

minima & maxima in x can be found from
[1M] $\frac{dx}{dt} = y = 0\ with\ \frac{dy}{dt} = -x - 2y \neq 0$: [1A] $y = 0$

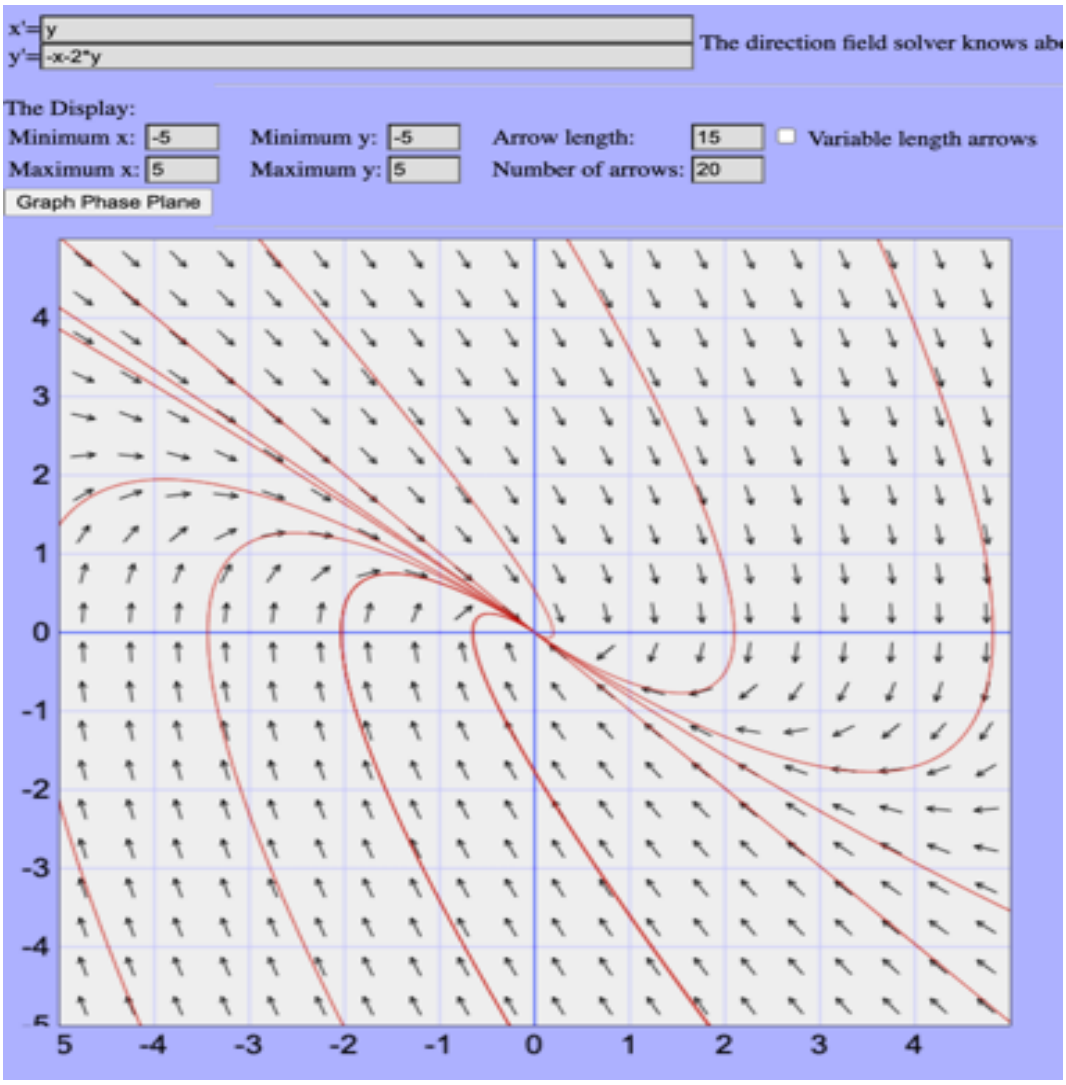
minima & maxima in y can be found from
[1M] $\frac{dy}{dt} = -x - 2y = 0 \quad with\ \frac{dy}{dt} = y \neq 0$: [1A]$y = -\frac{x}{2}$

- the orbit (or orbits) where the ratio of y/x remains constant for all time and the value of this ratio;

[1A] ratio of y/x remains constant for all time along straight line orbits ie eigenvectors.
y/x = -1

- the type and stability of the critical point at (0,0).

[1A+1A]Stable node



[1 axes labels]
See above for other sketch marks: total=6

QUESTION 2 [12 marks]

Use Laplace Transform to solve

$$x''(t) + 2x'(t) + x(t) = 0$$

$$x(0) = a, \quad x'(0) = b$$

$$\mathcal{L}\{x''^{(t)} + 2x'(t) + x(t)\} = 0$$

$\{s^2X - sx(0) - x'(0)\} + 2\{sX - x(0)\} + X = 0$ [1]

$$\{s^2X - sa - b\} + 2\{sX - a\} + X = 0$$

$(s^2 + 2s + 1)X = sa + 2a + b$ [1]

$X = \frac{sa+a+a+b}{(s+1)^2} = \frac{(s+1)a}{(s+1)^2} + \frac{a}{(s+1)^2} + \frac{b}{(s+1)^2}$ [1 for set up of expressions in form of LT Table]

$x(t) = ae^{-t} + (a+b)te^{-t}$ [1]

Find $\frac{x\prime(10)}{x(10)}$ given

$$x(0) = 1, \qquad x'(0) = -1$$

[Hint: you may use any relevant result (if such exist) from Q1]

$$x'(t) = -ae^{-t} - (a+b)te^{-t} + (a+b)e^{-t}$$

For

$$x(0) = a, \qquad x'(0) = b = -a$$

yields

$\frac{x\prime(t)}{x(t)} = -1, \forall t > 0.$ [1M+1A]

OR the student can simply use their results from Q1 to directly answer $\frac{x\prime(10)}{x(10)} = -1.$

Q2 continued

Find the Laplace Transform of

$$\int_0^t \delta(\tau - 2)d\tau$$

[2A]

$$\mathcal{L}\left\{\int_0^t \delta(\tau - 2)d\tau\right\} = \frac{e^{-2s}}{s}$$

Find y(t), given

$$\int_0^t \delta(\tau - 2)d\tau = 5\int_0^t y(\tau)d\tau + \frac{dy}{dt} + 2y, \qquad y(0) = 0$$

[1A+1A+1A+1A]

$$\mathcal{L}\left\{\int_0^t \delta(\tau - 2)d\tau\right\} = \mathcal{L}\left\{5\int_0^t y(\tau)d\tau + \frac{dy}{dt} + 2y\right\}, \qquad y(0) = 0$$

$$\frac{e^{-2s}}{s} = \frac{5Y}{s} + sY - y(0) + 2Y$$

[1Method isolating Y]

$$e^{-2s} = 5Y + s^2Y + 2sY = (s^2 + 2s + 5)Y$$

$$Y = \frac{1}{(s^2 + 2s + 5)}e^{-2s}$$

$$Y = \frac{1}{2}\frac{2}{((s+1)^2 + 2^2)}e^{-2s}$$

[1 for recognising t shift]

$$y = \frac{1}{2} f(t-2)u(t-2)$$

[1 for f(t)]

$$f(t) = \mathcal{L}^{-1}[F(s+1)] = e^{-t}\sin 2t$$

[1+1, 1 for each piece]

$$y = \begin{cases} 0, & 0 \le t \le 2 \\ \frac{1}{2} e^{-(t-2)} \sin 2(t-2), & t > 2 \end{cases}$$

QUESTION 3 [22 marks + 1 bonus mark]

Consider the Bessel's equation of order zero

$$r^2R'' + rR' + (\omega r)^2 R = 0$$

where $\omega > 0$.

(a) By trying a power series solution in the form

$$R(r) = \sum_{n=0}^{\infty} c_n r^n$$

derive recurrence relations for the coefficients $c_0, c_1, c_2, c_3, c_4, c_5$

(b)

(c) Deduce from part (a) a general expression for the coefficient c_n and thereby write down one linearly independent series solution for $R(r)$

Try the solution in the form

$$R(r) = \sum_{n=0}^{\infty} c_n r^n \qquad \dots (5)$$

$$R'(r) = \sum_{n=0}^{\infty} c_n n r^{n-1}$$

[1]

$$R''(r) = \sum_{n=0}^{\infty} c_n n(n-1) r^{n-2}$$

[1]

$$r^2 \sum_{n=0}^{\infty} c_n n(n-1) r^{n-2} + r \sum_{n=0}^{\infty} c_n n r^{n-1} + \omega^2 r^2 \sum_{n=0}^{\infty} c_n r^n = 0$$

$$\sum_{n=0}^{\infty} c_n n(n-1) r^n + \sum_{n=0}^{\infty} c_n n r^n + \omega^2 \sum_{n=0}^{\infty} c_n r^{n+2} = 0$$

[1]

Do index shift on 3rd term, by setting $m = n + 2 \;\; \rightarrow n = m - 2$, & replacing n with m in first two terms to get

$$\left[0 + 0 + \sum_{m=2}^{\infty} c_m\, m(m-1) r^m\right] + \left[0 + c_1 r + \sum_{m=2}^{\infty} c_m\, m r^m\right] + \left[\omega^2 \sum_{m=2}^{\infty} c_{m-2}\, r^m\right] = 0$$

[1]

Keep in mind the RHS is simply a series with all the coefficients equal to zero!

$$0 \;=\; 0r^0 + \;\; 0r^1 + \sum_{m=2}^{\infty} (0) r^m$$

Thus, by equating coefficients of like powers of r, we get

$$c_1 = 0$$

[1]

$$m(m-1)\, c_m + \; m\, c_m + \omega^2 c_{m-2} = 0, \quad m \geq 2$$

$$m^2\, c_m + \omega^2 c_{m-2} = 0, \quad m \geq 2$$

[1]

$$c_m = -\frac{\omega^2}{m^2} c_{m-2}\,, \quad m \geq 2$$

[1]

$$c_2 = -\frac{\omega^2}{2^2(1!)} c_0\,, \quad m = 2$$

[1]

$$c_3 = -\frac{\omega^2}{3^2} c_1 = 0, \quad m = 3$$

[1]

$$c_4 = -\frac{\omega^2}{4^2} c_2 = \frac{\omega^4}{4^2 2^2} c_0 = \frac{\omega^4}{(2^4)(2!)^2} c_0 \quad m = 4$$

$$c_5 = -\frac{\omega^2}{5^2} c_3 = \; 0, \quad m = 5$$

$$c_6 = -\frac{\omega^2}{6^2} c_4 = \; - \frac{\omega^2}{(3.2)^2} \frac{\omega^6}{(2^4)(2!)^2} c_0 \; = -\frac{\omega^6}{(2^6)(3!)^2} c_0, \quad m = 6$$

NOTE!!!!!

$(2.3)^2 (2^2)^2 2^2 = 2^6 (3!)^2$

$$c_{2n} = (-1)^n \frac{\omega^{2n}}{(2^{2n})(n!)^2} c_0, \quad m = 6 = 2n, \quad n = 1,2,3$$

$$\ldots etc$$

[1]

$$R(r) = c_0 \left\{ 1 + 0(\omega r) + \frac{(-1)}{2^2(1!)}(\omega r)^2 + 0(\omega r)^3 + \frac{(-1)^2}{(2^4)(2!)^2}(\omega r)^4 + 0(\omega r)^5 + \frac{(-1)^3}{(2^6)(3!)^2}(\omega r)^6 + \cdots \right\}$$

$$\boxed{r^2 R'' + rR' + (\omega r)^2 R = 0}$$

$$R(r) = c_0 \left\{ \sum_{n=0}^{\infty} (-1)^n \frac{\omega^{2n}}{(2^{2n})(n!)^2} \quad r^{2n} \right\} = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2^{2n})(n!)^2} \quad (\omega r)^{2n} = c_0 J_0(\omega r)$$

Substituting (6) into (5) yields one linearly independent series solution

[1]

$$r^2 R'' + rR' + (\omega r)^2 R = 0 \qquad \ldots \ (1)$$

Try

$$R(r) = v(r) J_0(\omega r) \qquad \ldots (2A)$$

$$R' = \frac{d[v(r)J_0(\omega r)]}{dr} = \frac{dv}{dr} J_0(\omega r) + v \frac{dJ_0(\omega r)}{dr} \ \ldots (2B)$$

$$R'' = \frac{d}{dr}\left\{ \frac{d[v(r)J_0(\omega r)]}{dr} \right\} = \frac{d}{dr}\left\{ \frac{dv}{dr} J_0(\omega r) + v \frac{dJ_0(\omega r)}{dr} \right\}$$

$$= \frac{d^2v}{dr^2}J_0(\omega r) + \frac{dv}{dr}\frac{dJ_0(\omega r)}{dr} + \frac{dv}{dr}\frac{dJ_0(\omega r)}{dr} + v\frac{d^2J_0(\omega r)}{dr^2}$$

$$= \frac{d^2v}{dr^2}J_0(\omega r) + 2\frac{dv}{dr}\frac{dJ_0(\omega r)}{dr} + v\frac{d^2J_0(\omega r)}{dr^2} \quad \ldots(2C)$$

Substitute (2A-2C) into (1) to get

$$r^2\left(\frac{d^2v}{dr^2}J_0(\omega r) + 2\frac{dv}{dr}\frac{dJ_0(\omega r)}{dr} + v\frac{d^2J_0(\omega r)}{dr^2}\right) + r\left(\frac{dv}{dr}J_0(\omega r) + v\frac{dJ_0(\omega r)}{dr}\right) + (\omega r)^2 vJ_0 = 0 \quad \ldots(3)$$

[1]

Keeping in mind that since $J_0 = J_0(\omega r)$ is a solution to (1), we know that

$$r^2 v\frac{d^2J_0(\omega r)}{dr^2} + rv\frac{dJ_0(\omega r)}{dr} + (\omega r)^2 vJ_0 = 0,$$

Thus Eqn (3) reduces to

$$r^2\left(\frac{d^2v}{dr^2}J_0(\omega r) + 2\frac{dv}{dr}\frac{dJ_0(\omega r)}{dr}\right) + r\frac{dv}{dr}J_0(\omega r) = 0$$

$$\frac{d^2v}{dr^2} + \left(\frac{2}{J_0(\omega r)}\frac{dJ_0(\omega r)}{dr} + \frac{1}{r}\right)\frac{dv}{dr} = 0 \quad \ldots(4)$$

[1]

Eqn (4) is a first order eqn in $\frac{dv}{dr}$.

The integrating factor is

$$I(r) = \exp\left(2\int\frac{J_0{}'}{J_0}dr + \int\frac{1}{r}dr\right) = \exp\left(2\int\frac{J_0{}'}{J_0}dr\right)\exp\left(\int\frac{1}{r}dr\right)$$

$$I(r) = e^{2\ln J_0}e^{\ln r} = rJ_0{}^2$$

[1 for working+1 answer]

Hence

$$\frac{d}{dr}\left\{\frac{dv}{dr}rJ_0{}^2\right\} = 0 \qquad \rightarrow \frac{dv}{dr} = \frac{D}{rJ_0{}^2}$$

$$v = \int\frac{dv}{dr}dr + C$$

$$v = \int \frac{D}{r{J_0}^2} dr \ + \ C$$

where D & C are real constants.

[1 for working+1 answer]

Recalling (2), we get

$$R(r) = DJ_0(\omega r) \int \frac{1}{r(J_0(\omega r))^2} dr \ + \ CJ_0(\omega r)$$

[1]

Note that as $r \to 0$, J_0 approaches a constant and therefore there is a logarithmic singularity at $r = 0$.
[1 bonus mark for saying something about the singularity]

$$R(r) = CJ_0(\omega r) + \ D[2nd\ lin\ indep\ soln]; \qquad C, D \ \in \mathbb{R} \quad \ldots (7)$$

QUESTION 4 [25 marks]

(a) Consider the function

$$f(x) = \frac{1}{2}, \quad \text{for} \quad 0 < x < 1, \quad f(0) = f(1) = 0, \quad f(-x) = -f(x), \quad f(x+2) = f(x).$$

(i) Is the function $f(x)$ even, odd or neither?

$f(x)$ is odd.

[2]

(ii) Calculate the Fourier series of $f(x)$.
Hint: only calculate the non-zero coefficients based on your answer in (i).

Because $f(x)$ is odd, $a_0 = a_n = 0$ and

$$b_n = 2\int_0^1 \frac{1}{2}\sin n\pi x dx$$
$$= -\left[\frac{1}{n\pi}\cos n\pi x\right]_0^1$$
$$= -\frac{(-1)^n - 1}{n\pi}$$

Now for $n = 2m$, $b_{2m} = 0$. For $n = 2m-1$,

$$b_{2m-1} = \frac{2}{(2m-1)\pi}$$

and so the Fourier series of $f(x)$ is

$$s(x) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{2m-1}\sin\big([2m-1]\pi x\big)$$

[1A] $a_n = 0$ either from calculating or reasoning

[1M] Calculating b_n

[1A] Correct result before simplification

[1M] Even terms zero

[1A] Correctly simplified b_{2m-1}

[1A] Correct F.S. with or without simplification

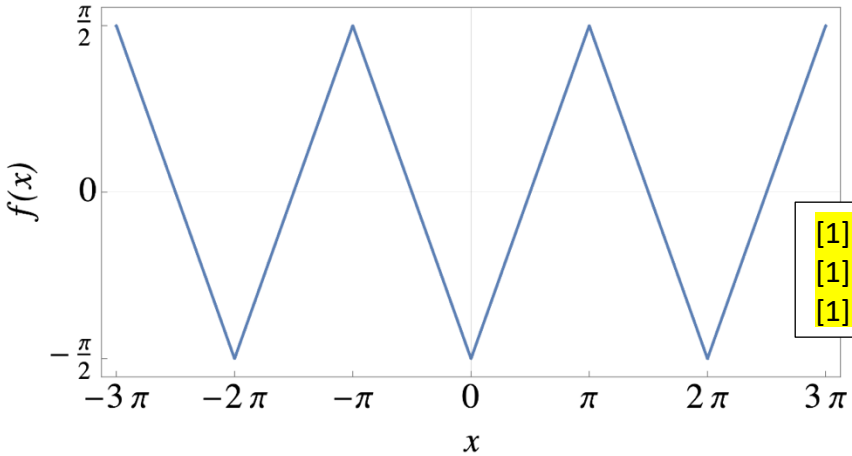
(b) Consider the function

$$f(x) = x - \frac{\pi}{2} \qquad \text{for} \qquad x \in [0, \pi], \qquad f(-x) = f(x), \qquad f(x + 2\pi) = f(x).$$

The Fourier series of $f(x)$, where $L = \pi$, is

$$s(x) = -\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2m-1)^2} \cos\big([2m-1]x\big).$$

(i) Sketch $f(x)$ for $x \in [-3\pi, 3\pi]$.



[1] Between $[0, \pi]$ correct
[1] Between $[-3\pi, 3\pi]$ correct
[1] Labelled correctly

(ii) Use the pointwise convergence theorem, and $s(x)$, to evaluate the sum

$$\sum_{m=1}^{\infty} \frac{1}{(2m-1)^2}.$$

At $x = 0$, $f(0) = -\frac{\pi}{2}$. Since $\cos([2m-1] \times 0) = 1$, the pointwise convergence theorem gives us

$$-\frac{\pi}{2} = -\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2m-1)^2}$$

which upon rearrangement gives the required result

$$\sum_{n=1}^{\infty} \frac{1}{(2m-1)^2} = \frac{\pi^2}{8}$$

[1M] Using point $x = 0$
[1M] Substituting into Fourier series
[1A] Final answer

(iii) Does the Fourier series of $f(x)$ exhibit Gibbs phenomenon? If so, what is the error of the truncated Fourier series approximation at this point?

$f(x)$ is continuous over $x \in \mathbb{R}$ and the derivative of $f(x)$ is piecewise continuous so the Fourier series of $f(x)$ will NOT exhibit Gibbs phemonenon.

[2] No Gibbs

(iv) Does the Fourier series of $f(x)$ converge uniformly? If so, prove this using the Weierstrass test for uniform convergence.

[1] Converges uniformly

For the same reason as (c), the Fourier series of $f(x)$ DOES converge uniformly. To use the Weierstrass test for uniform convergence we must find something which bounds each term of the Fourier series, i.e.,

[1M] Attempting to find bound
[1A] Correct bound

$$\left| -\frac{4}{\pi} \frac{1}{(2m-1)^2} \cos\big([2m-1]x\big) \right| \leq \frac{4}{\pi} \frac{1}{(2m-1)^2}$$

for all $x \in \mathbb{R}$ and $m \geq 1$. Now application of the pointwise convergence theorem from (b) tells us that $\displaystyle\sum_{m=1}^{\infty} \frac{1}{(2m-1)^2}$ converges and so we have found a bound that converges. Thus the Weierstrass test for uniform convergence tells us that the Fourier series converges uniformly.

There are many other alternative series we could have chosen. For example the series $\displaystyle\frac{4}{\pi} \sum_{m=1}^{\infty} \frac{1}{m^2}$ would also work in the Weierstrass test.

[1A] Overall convincing use of Weierstrass test

(v) Find an antiderivative of $s(x)$ using

$$S(x) = \int_0^x s(\xi)d\xi$$

and term by term integration.

$$\begin{aligned} S(x) &= -\frac{4}{\pi}\sum_{m=1}^{\infty}\int_0^x \frac{1}{(2m-1)^2}\cos\big([2m-1]\xi\big)d\xi \\ &= -\frac{4}{\pi}\sum_{m=1}^{\infty}\left[\frac{1}{(2m-1)^3}\sin\big([2m-1]\xi\big)\right]_0^x \\ &= -\frac{4}{\pi}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^3}\sin\big([2m-1]x\big) \end{aligned}$$

[1M] Integrating term-by-term

[1M]

[1A] Final answer

(vi) Is the term-by-term integration in (e) justified? Explain your answer in one sentence.

Because the Fourier series of $f(x)$ converges uniformly, we are allowed to integrate the sum term by term and the result will also be uniformly convergent.

[1] Yes
[1] Justification

QUESTION 5 [19 marks + 2BONUS]

Part 5a

[1M for separating variables]
Substitute (2) into (1) to get

$$\frac{\partial^2 RT}{\partial t^2} = \frac{\partial^2 RT}{\partial r^2} + \frac{1}{r}\frac{\partial RT}{\partial r}$$

$$RT'' = R''T + \frac{1}{r}TR'$$

Then obtain a separated form to see if amenable to SoV

$$\frac{RT''}{RT} = \frac{R''T}{RT} + \frac{1}{r}\frac{TR'}{RT}$$

[1 for showing separated form is possible only if each side of the equation equals the same constant]

$$\frac{T''}{T} = \frac{R''}{R} + \frac{1}{r}\frac{R'}{R} = \quad -\lambda$$

[1+1 for two ODEs]

$$r^2R'' + rR' \ + \ \lambda r^2 R = 0 \qquad \ldots \ (3A)$$

$$T'' + \ \lambda T = 0 \qquad \ldots \ (3B)$$

Part 5b

[1]
Case 1 ($\lambda > 0$): this leads to oscillatory vibrations in time
Case 2 ($\lambda = 0$)
Case 3 ($\lambda < 0$)

Part 5ci

[1 for ODE in R]

$$\lambda = \omega^2, \quad \omega > 0$$
$$r^2R'' + rR' + \omega^2 r^2 R = 0$$

[1 for BC at drum edge]
Boundary condition $u(a,t) = 0, \quad \forall t > 0$ leads to

$$u(a,t) = R(a)T(t) = 0, \quad \forall t > 0 \qquad \rightarrow R(a) = 0$$

[1 for BC at drum centre]
$R(a) = 0, \quad R(0)$ must remain finite for $\forall t > 0$.

[1 for solution to R]

$$R_{general}(r) = CJ_0(\omega r) + D[Y_0(\omega r)]; \qquad C, D \in \mathbb{R}$$

[1 for applying BC at drum centre]
Since $R(0)$ must remain finite for $\forall t > 0$, $D = 0$

$$R(r) = CJ_0(\omega r)$$

[1 for applying BC at drum edge]
Since $R(a) = 0$ we get

$$R(a) = 0 \quad \rightarrow CJ_0(\omega a) = 0$$
$$J_0(\omega a) = 0$$

[1 for recognising there are an infinite number of zeroes for J_0]
Call these zeroes $\bar{\omega}_n$, $n = 1,2,3, \ldots$ such that $\bar{\omega}_n = \frac{\omega_n}{a}$

[1 for final solution R_n]

$$R_n(r) = CJ_0(\bar{\omega}_n r) = CJ_0\left(\frac{\omega_n}{a} r\right)$$

Part 5cii

[1 for Solving the corresponding ODE in $T_n(t)$]
$T_n(t) = A_n sin(\bar{\omega}_n t) + B_n cos(\bar{\omega}_n t)$ or

$$T_n(t) = A_n sin\left(\frac{\omega_n}{a} t\right) + B_n cos\left(\frac{\omega_n}{a} t\right)$$

Part 5ciii

[1 for the general solution]

$$u(r,t) = \sum_{n=1}^{\infty} [J_0(\bar{\omega}_n r)]\{A_n sin(\bar{\omega}_n t) + B_n cos(\bar{\omega}_n t)\}$$

[1 for differentiation]

$$u_t(r,t) = \sum_{n=1}^{\infty} [J_0(\bar{\omega}_n r)]\{A_n \bar{\omega}_n cos(\bar{\omega}_n t) - B_n \bar{\omega}_n sin(\bar{\omega}_n t)\}$$

[1 mark for interpreting "released from rest" with 1 bonus for writing $\forall\, r \in [0,a]$]

$$u_t(r,0) = 0 \quad \forall\, r \in [0,a]$$

Award only 1 mark if student just writes $u_t(r,0) = 0$

[1 for answer to coefficients A_n]

$$u_t(r,0) = \sum_{n=1}^{\infty} [J_0(\bar{\omega}_n r)]\{A_n \bar{\omega}_n\} = 0$$

$$A_n = 0, \qquad \forall n \geq 1$$

[BONUS of 1 for providing justification to answer]

Since

$$\bar{\omega}_n > 0, \qquad \forall n \geq 1$$

and $J_0(\bar{\omega}_n r)$ is only zero at zeroes of J_0 and not zero $\forall\, r \in [0,a]$.

[1 for final solution]

$$u(r,t) = \sum_{n=1}^{\infty} B_n J_0(\bar{\omega}_n r) cos(\bar{\omega}_n t)$$

QUESTION 6 [11 marks+1BONUS]

Part (ai)

[1 mark for integral formula for total heat] = area under the curve
[1 mark for working out integral or clever student would simply use area of a triangle]
[1 mark for answer]

$$Total\ Heat = \int_0^1 u(x,0)dx = area\ of\ triangle = \frac{1}{2}(1)\left(\frac{1}{2}\right) = \frac{1}{4}$$

OR

$$Total\ Heat = \int_0^1 u(x,0)dx = \int_0^{1/2} xdx + \int_{1/2}^1 (1-x)dx = \left[\frac{x^2}{2}\right]_0^{1/2} + \left[x - \frac{x^2}{2}\right]_{1/2}^1$$
$$= \frac{1}{8} + \left(\frac{1}{2}\right) - \left(\frac{1}{2} - \frac{1}{8}\right) = \frac{1}{4}$$

[1 mark for answer:
Steady state temperature is

$$\lim_{t\to\infty} u(x,t) = \frac{1}{4}$$

[1 mark for Justification: the rod is 100% insulated, so there is no heat loss]

$$\int_0^1 \left(\lim_{t\to\infty} u(x,t)\right) dx = \frac{1}{4}$$

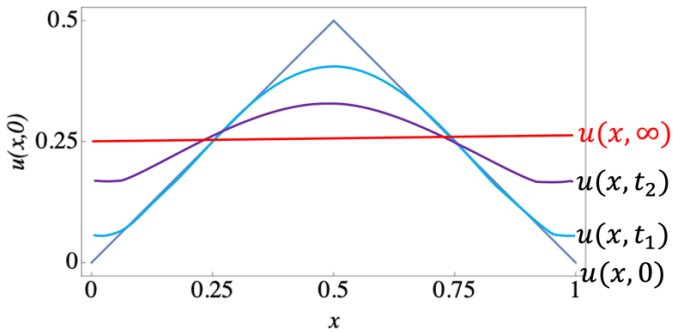
[1mark for Justification: the steady state temperature must be uniform along the rod]

$$\lim_{t\to\infty} u(x,t) = constant$$

Part (aii)

[3marks total: 1 mark for each profile for t>0 but award only if shape has all the correct features:

- zero slope at boundaries-so check curves are flat here
- symmetric and decreasing area under the curve, with previously cool (warm) areas getting warmer (cooler)]



Part (bi)

[1 mark] for "heat eqn is a smoothing operation and thus smooths out the discontinuity"

[1 mark + **1 bonus**] for "wave eqn propagates the discontinuity **periodically** in time"

QUESTION 7 [7 marks]

Consider the ODE

$$\frac{d^5y}{dx^5} + xy = 0$$

on the open domain $x \in \mathbb{R}$.

Use the method of Fourier transforms to find the most general solution to the ODE that has a Fourier transform. You may leave your final answer as a definite integral.

Transforming the ODE into Fourier space gives

$$\mathscr{F}\left\{\frac{d^5y}{dx^5}\right\} + \mathscr{F}\{xy\} = 0$$

$$i^5k^5\widehat{y} + i\frac{d\widehat{y}}{dk} = 0$$

$$\frac{d\widehat{y}}{dk} - k^5\widehat{y} = 0.$$

[1M] Move to Fourier space

[1M] Simplify ODE

The integrating factor is therefore $I = e^{\int -k^5 dk} = e^{-\frac{k^6}{6}}$ and so

$$\frac{d}{dk}\left(e^{-\frac{k^6}{6}}\widehat{y}\right) = 0$$

$$e^{-\frac{k^6}{6}}\widehat{y} = C$$

$$\widehat{y} = Ce^{\frac{k^6}{6}}.$$

[1A] Correct integrating factor

[1M] Solve ODE for $\hat{y}$

[1A] Solution for $\hat{y}$

Applying the formula for inverting the Fourier transform then gives the most general solution which has a Fourier transform:

$$y = D\int_{-\infty}^{\infty} e^{\frac{k^6}{6}+ikx}dk$$

where D is a new arbitrary constant ($D = 2\pi C$).

[1M] Inverting Fourier transform

[1A] Final answer